

Gini Index Decision Tree for Play Tennis

Pen-and-paper method with full calculations

Target classes: **Yes** = Play Tennis, **No** = Do Not Play. In Gini-based decision tree construction, we select the attribute with the **minimum weighted Gini** after splitting.

Play Tennis Dataset

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Formula Used

For any set S with class probabilities $p_1, p_2, \dots, p_n$:

$$\text{Gini}(S) = 1 - \sum (p_i^2)$$

For a split on attribute A, compute the weighted Gini of all child subsets:

$$\text{Weighted Gini}(A) = \sum (|S_v| / |S|) * \text{Gini}(S_v)$$

Choose the attribute that gives the **smallest weighted Gini**.

Step 1: Gini of the whole dataset

Class counts in the full dataset:

$$\text{Yes} = 9, \text{ No} = 5, \text{ Total} = 14$$

$$\begin{aligned}\text{Gini}(S) &= 1 - (9/14)^2 - (5/14)^2 \\ &= 1 - 81/196 - 25/196 \\ &= 1 - 106/196 \\ &= 90/196\end{aligned}$$

$$= 0.459$$

Therefore, the Gini of the complete dataset is **0.459**.

Step 2: Weighted Gini for each attribute

A. Attribute: Outlook

Sunny subset: 5 rows -> Yes = 2, No = 3

$$\begin{aligned} \text{Gini(Sunny)} &= 1 - (2/5)^2 - (3/5)^2 \\ &= 1 - 4/25 - 9/25 \\ &= 12/25 \\ &= 0.48 \end{aligned}$$

Overcast subset: 4 rows -> Yes = 4, No = 0

$$\begin{aligned} \text{Gini(Overcast)} &= 1 - (4/4)^2 - (0/4)^2 \\ &= 0 \end{aligned}$$

Rain subset: 5 rows -> Yes = 3, No = 2

$$\begin{aligned} \text{Gini(Rain)} &= 1 - (3/5)^2 - (2/5)^2 \\ &= 1 - 9/25 - 4/25 \\ &= 12/25 \\ &= 0.48 \end{aligned}$$

$$\begin{aligned} \text{Weighted Gini(Outlook)} &= (5/14)*0.48 + (4/14)*0 + (5/14)*0.48 \\ &= 0.171 + 0 + 0.171 \\ &= 0.343 \end{aligned}$$

B. Attribute: Temperature

Hot subset: 4 rows -> Yes = 2, No = 2

$$\begin{aligned} \text{Gini(Hot)} &= 1 - (2/4)^2 - (2/4)^2 \\ &= 1 - 1/4 - 1/4 \\ &= 0.5 \end{aligned}$$

Mild subset: 6 rows -> Yes = 4, No = 2

$$\begin{aligned} \text{Gini(Mild)} &= 1 - (4/6)^2 - (2/6)^2 \\ &= 1 - 16/36 - 4/36 \\ &= 16/36 \\ &= 0.444 \end{aligned}$$

Cool subset: 4 rows -> Yes = 3, No = 1

$$\begin{aligned} \text{Gini(Cool)} &= 1 - (3/4)^2 - (1/4)^2 \\ &= 1 - 9/16 - 1/16 \\ &= 6/16 \\ &= 0.375 \end{aligned}$$

$$\begin{aligned} \text{Weighted Gini(Temperature)} &= (4/14)*0.5 + (6/14)*0.444 + (4/14)*0.375 \\ &= 0.143 + 0.190 + 0.107 \\ &= 0.440 \end{aligned}$$

C. Attribute: Humidity

High subset: 7 rows -> Yes = 3, No = 4

$$\begin{aligned} \text{Gini(High)} &= 1 - (3/7)^2 - (4/7)^2 \\ &= 1 - 9/49 - 16/49 \\ &= 24/49 \\ &= 0.490 \end{aligned}$$

Normal subset: 7 rows -> Yes = 6, No = 1

$$\begin{aligned} \text{Gini(Normal)} &= 1 - (6/7)^2 - (1/7)^2 \\ &= 1 - 36/49 - 1/49 \end{aligned}$$

$$= 12/49$$

$$= 0.245$$

$$\text{Weighted Gini(Humidity)} = (7/14)*0.490 + (7/14)*0.245$$

$$= 0.245 + 0.123$$

$$= 0.368$$

D. Attribute: Wind

Weak subset: 8 rows -> Yes = 6, No = 2

$$\text{Gini(Weak)} = 1 - (6/8)^2 - (2/8)^2$$

$$= 1 - 36/64 - 4/64$$

$$= 24/64$$

$$= 0.375$$

Strong subset: 6 rows -> Yes = 3, No = 3

$$\text{Gini(Strong)} = 1 - (3/6)^2 - (3/6)^2$$

$$= 1 - 1/4 - 1/4$$

$$= 0.5$$

$$\text{Weighted Gini(Wind)} = (8/14)*0.375 + (6/14)*0.5$$

$$= 0.214 + 0.214$$

$$= 0.428$$

Step 3: Select the root node

Attribute	Weighted Gini
Outlook	0.343
Humidity	0.368
Wind	0.428
Temperature	0.440

The minimum weighted Gini is **0.343**. Therefore, the root node is **Outlook**.

Step 4: Split each branch

Branch 1: Outlook = Overcast

Rows: 3, 7, 12, 13
 Yes = 4, No = 0
 Gini = 0

This node is pure, so:

Overcast -> Yes

Branch 2: Outlook = Sunny

Sunny subset rows: 1, 2, 8, 9, 11

Try split on Humidity:

Humidity = High -> rows 1, 2, 8 -> Yes = 0, No = 3
 Gini(High) = 0

Humidity = Normal -> rows 9, 11 -> Yes = 2, No = 0
 Gini(Normal) = 0

$$\text{Weighted Gini} = (3/5)*0 + (2/5)*0 = 0$$

So Humidity perfectly splits the Sunny branch:

Sunny

- Humidity = High -> No
- Humidity = Normal -> Yes

Branch 3: Outlook = Rain

Rain subset rows: 4, 5, 6, 10, 14

Try split on Wind:

Wind = Weak -> rows 4, 5, 10 -> Yes = 3, No = 0
 Gini(Weak) = 0

Wind = Strong -> rows 6, 14 -> Yes = 0, No = 2
 Gini(Strong) = 0

$$\text{Weighted Gini} = (3/5)*0 + (2/5)*0 = 0$$

So Wind perfectly splits the Rain branch:

Rain

- Wind = Weak -> Yes
- Wind = Strong -> No

Final Decision Tree

Outlook

```
|-- Overcast -> Yes
|-- Sunny
|   |-- Humidity
|       |-- High -> No
|       |-- Normal -> Yes
|-- Rain
    |-- Wind
        |-- Weak -> Yes
        |-- Strong -> No
```

Final Rules

1. If Outlook = Overcast, then PlayTennis = Yes
2. If Outlook = Sunny and Humidity = High, then PlayTennis = No
3. If Outlook = Sunny and Humidity = Normal, then PlayTennis = Yes
4. If Outlook = Rain and Wind = Weak, then PlayTennis = Yes
5. If Outlook = Rain and Wind = Strong, then PlayTennis = No

Conclusion: In the Gini method, the best attribute is the one with the lowest weighted impurity after splitting. For the Play Tennis dataset, the root node is Outlook.